

* Rank 0 tensor.

$$[Q]$$

$$Q = Q_0^{(1)}$$

* Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} Q_{xx} & 0 & Q_{xz} \\ 0 & Q_{yy} & 0 \\ Q_{xz} & 0 & Q_{zz} \end{bmatrix}$$

$$Q_{xx} = Q_0^{(1)} - Q_u^{(1)} + Q_v^{(1)}$$

$$Q_{xz} = Q_{xz}^{(1)}$$

$$Q_{yy} = Q_0^{(1)} - Q_u^{(1)} - Q_v^{(1)}$$

$$Q_{zz} = Q_0^{(1)} + 2Q_u^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & Q_{xz} \\ 0 & 0 & 0 \\ -Q_{xz} & 0 & 0 \end{bmatrix}$$

$$Q_{xz} = -G_y^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ Q_{xxyy} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ Q_{xxzz} & Q_{yyzz} & Q_{zzzz} & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{xxzx} & Q_{yyzx} & Q_{zzzx} & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & Q_{yzxy} & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{xxxx} = Q_0^{(1)} + 2Q_0^{(2)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} + 2Q_4^{(1)} - 2Q_u^{(1)} - 4Q_u^{(2)} + 2Q_v^{(1)} + 4Q_v^{(2)}$$

$$Q_{xxyy} = Q_0^{(1)} + 2Q_{4u}^{(1)} - Q_4^{(1)} - 2Q_u^{(1)}$$

$$Q_{xxzz} = Q_0^{(1)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(1)} + Q_v^{(1)}$$

$$Q_{xxzx} = -Q_{4y}^{\alpha(1)} + Q_{xz}^{(1)} + 2Q_{xz}^{(2)} - Q_y^{\beta(1)}$$

$$Q_{yyyy} = Q_0^{(1)} + 2Q_0^{(2)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} + 2Q_4^{(1)} - 2Q_u^{(1)} - 4Q_u^{(2)} - 2Q_v^{(1)} - 4Q_v^{(2)}$$

$$Q_{yyzz} = Q_0^{(1)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(1)} - Q_v^{(1)}$$

$$Q_{yyzx} = Q_{xz}^{(1)} + 2Q_y^{\beta(1)}$$

$$Q_{zzzz} = Q_0^{(1)} + 2Q_0^{(2)} + 2Q_{4u}^{(1)} + 2Q_4^{(1)} + 4Q_u^{(1)} + 8Q_u^{(2)}$$

$$Q_{zzzx} = Q_{4y}^{\alpha(1)} + Q_{xz}^{(1)} + 2Q_{xz}^{(2)} - Q_y^{\beta(1)}$$

$$Q_{yzyz} = Q_0^{(2)} - Q_{4u}^{(1)} + Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(2)} - Q_v^{(2)}$$

$$Q_{yzxy} = Q_{xz}^{(2)} + 2Q_y^{\beta(1)}$$

$$Q_{zxzx} = Q_0^{(2)} - Q_{4u}^{(1)} - Q_{4v}^{(1)} - Q_4^{(1)} + Q_u^{(2)} + Q_v^{(2)}$$

$$Q_{xyxy} = Q_0^{(2)} + 2Q_{4u}^{(1)} - Q_4^{(1)} - 2Q_u^{(2)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxzx} & 0 \\ -Q_{xxyy} & 0 & Q_{yyzz} & 0 & Q_{yyzx} & 0 \\ -Q_{xxzz} & -Q_{yyzz} & 0 & 0 & Q_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & Q_{yzxy} \\ -Q_{xxzx} & -Q_{yyzx} & -Q_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -Q_{yzxy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} Q_{xxyy} &= 4\mathbb{G}_{xyz}^{(1)} - 2\mathbb{Q}_v^{(3)} \\ Q_{xxzz} &= -4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(3)} \\ Q_{xxzx} &= -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)} \\ Q_{yyzz} &= 4\mathbb{G}_{xyz}^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(3)} \\ Q_{yyzx} &= -4\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)} \\ Q_{zzzx} &= 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(3)} \\ Q_{yzxy} &= \mathbb{G}_y^{(1)} + 4\mathbb{G}_y^{\alpha(1)} \end{aligned}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} Q_{yzyz} & 0 & Q_{yzxy} \\ 0 & Q_{zxzx} & 0 \\ Q_{yzxy} & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{yzyz} &= \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(4)} \\ Q_{yzxy} &= 2\mathbb{Q}_{xz}^{(4)} \\ Q_{zxzx} &= \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 2\mathbb{Q}_v^{(4)} \\ Q_{xyxy} &= \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} \end{aligned}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & Q_{yzxy} \\ 0 & 0 & 0 \\ -Q_{yzxy} & 0 & 0 \end{bmatrix}$$

$$Q_{yzxy} = \mathbb{G}_y^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & Q_{xxzx} & 0 \\ 0 & Q_{yyzx} & 0 \\ 0 & Q_{zzzx} & 0 \\ Q_{yzyz} & 0 & Q_{yzxy} \\ 0 & Q_{zxzx} & 0 \\ Q_{xyyz} & 0 & Q_{xyxy} \end{bmatrix}$$

$$\begin{aligned} Q_{xxzx} &= \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + 2\mathbb{Q}_{xz}^{(5)} \\ Q_{yyzx} &= \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} \\ Q_{zzzx} &= \mathbb{G}_y^{(3)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} - 2\mathbb{Q}_{xz}^{(5)} \\ Q_{yzyz} &= \mathbb{G}_{xyz}^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)} \\ Q_{yzxy} &= \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(5)} \\ Q_{zxzx} &= \mathbb{G}_{xyz}^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(5)} \\ Q_{xyyz} &= \mathbb{G}_y^{(4)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{xz}^{(5)} \\ Q_{xyxy} &= \mathbb{G}_{xyz}^{(2)} + 2\mathbb{Q}_v^{(5)} \end{aligned}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxxz} & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} \end{bmatrix}$$

$$Q_{yzyz} = \mathbb{G}_{xyz}^{(3)} - 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)}$$

$$Q_{yzxy} = \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - \mathbb{Q}_{xz}^{(6)}$$

$$Q_{zxxx} = \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)}$$

$$Q_{zxyy} = \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)}$$

$$Q_{zxxz} = \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)}$$

$$Q_{zxzx} = \mathbb{G}_{xyz}^{(3)} + 3\mathbb{Q}_u^{(6)} - \mathbb{Q}_v^{(6)}$$

$$Q_{xyyz} = \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + \mathbb{Q}_{xz}^{(6)}$$

$$Q_{xyxy} = \mathbb{G}_{xyz}^{(3)} + 2\mathbb{Q}_v^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} Q_{xxxx} & Q_{xxyy} & Q_{xxzz} & 0 & Q_{xxxx} & 0 & 0 & Q_{xxxz} & 0 \\ Q_{yyxx} & Q_{yyyy} & Q_{yyzz} & 0 & Q_{yyyz} & 0 & 0 & Q_{yyxz} & 0 \\ Q_{zzxx} & Q_{zzyy} & Q_{zzzz} & 0 & Q_{zzzx} & 0 & 0 & Q_{zzxz} & 0 \\ 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\ Q_{zxxx} & Q_{zxyy} & Q_{zxxz} & 0 & Q_{zxxz} & 0 & 0 & Q_{zxxz} & 0 \\ 0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
Q_{xxxx} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)} \\
Q_{xxyy} &= 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)} \\
Q_{xxzz} &= -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)} \\
Q_{xxzx} &= -2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{xxxz} &= -2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} - \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{yyxx} &= -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} + 2\mathbb{Q}_v^{(3)} \\
Q_{yyyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)} \\
Q_{yyzz} &= 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)} \\
Q_{yyzx} &= \mathbb{G}_y^{(3)} + 2\mathbb{G}_y^{(4)} + 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{yyxz} &= -\mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{(4)} - 2\mathbb{G}_y^{\alpha(2)} - 4\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{zzxx} &= 4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)} \\
Q_{zzyy} &= -4\mathbb{G}_{xyz}^{(1)} + \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} - \mathbb{Q}_v^{(3)} \\
Q_{zzzz} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\
Q_{zzzx} &= 2\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{zzxz} &= 2\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(3)} - 2\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + 2\mathbb{G}_y^{\beta(1)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{yzyz} &= \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)} \\
Q_{yzxy} &= \mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{yzzy} &= -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)} \\
Q_{yzyx} &= \mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} + 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{zxzx} &= 2\mathbb{G}_y^{(1)} - 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} - \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{zxyy} &= 4\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{xz}^{(1)} - \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{zxzz} &= -2\mathbb{G}_y^{(1)} + 2\mathbb{G}_y^{\alpha(1)} - 2\mathbb{G}_y^{\beta(1)} + \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{zzxz} &= \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} - \mathbb{Q}_v^{(5)} \\
Q_{zzxx} &= -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)} \\
Q_{xyyz} &= -\mathbb{G}_y^{(1)} + \mathbb{G}_y^{(4)} - 4\mathbb{G}_y^{\alpha(1)} - \mathbb{G}_y^{\alpha(2)} - \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(5)} \\
Q_{xyzy} &= -\mathbb{G}_y^{(1)} - \mathbb{G}_y^{(4)} - 4\mathbb{G}_y^{\alpha(1)} + \mathbb{G}_y^{\alpha(2)} + \mathbb{G}_y^{\beta(2)} + \mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(5)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{xyyx} &= -\mathbb{G}_{xyz}^{(2)} + \mathbb{Q}_0^{(2)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(5)}
\end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & Q_{yzyz} & 0 & Q_{yzxy} & Q_{yzzy} & 0 & Q_{yzyx} \\ Q_{zxzx} & Q_{zxyy} & Q_{zzzz} & 0 & Q_{zzxx} & 0 & 0 & Q_{zxzx} & 0 \\ 0 & 0 & 0 & Q_{xyyz} & 0 & Q_{xyxy} & Q_{xyzy} & 0 & Q_{xyyx} \end{bmatrix}$$

$$\begin{aligned}
Q_{yzyz} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzxy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{yzzy} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} - 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{yzyx} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} - \mathbb{Q}_{xz}^{(6)} \\
Q_{zxxx} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} - \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zxyy} &= \mathbb{G}_y^{(5)} + 2\mathbb{G}_y^{(6)} + 2\mathbb{G}_y^{\alpha(3)} \\
Q_{zxzz} &= \mathbb{G}_y^{(5)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(6)} \\
Q_{zzxx} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} - 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} - 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{zzxz} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} + 2\mathbb{Q}_u^{(4)} + 3\mathbb{Q}_u^{(6)} + 2\mathbb{Q}_v^{(4)} - \mathbb{Q}_v^{(6)} \\
Q_{xyyz} &= -\mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} + 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyxy} &= \mathbb{G}_{xyz}^{(3)} + \mathbb{Q}_0^{(3)} + 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)} \\
Q_{xyzy} &= \mathbb{G}_y^{(2)} + \mathbb{G}_y^{(6)} - \mathbb{G}_y^{\alpha(3)} + \mathbb{G}_y^{\beta(3)} - 2\mathbb{Q}_{xz}^{(4)} + \mathbb{Q}_{xz}^{(6)} \\
Q_{xyyx} &= \mathbb{G}_{xyz}^{(3)} - \mathbb{Q}_0^{(3)} - 4\mathbb{Q}_u^{(4)} + 2\mathbb{Q}_v^{(6)}
\end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix}
Q_{xxxx} & 0 & Q_{xxxx} \\
0 & Q_{yyyy} & 0 \\
Q_{zzzz} & 0 & Q_{zzzz} \\
Q_{yyzz} & 0 & Q_{yyzz} \\
Q_{zzxx} & 0 & Q_{zzxx} \\
0 & Q_{xxyy} & 0 \\
0 & Q_{yzzz} & 0 \\
Q_{xxxy} + Q_{yyzz} - Q_{yzzz} - Q_{zzxz} + Q_{zzxx} & 0 & Q_{zzxx} \\
0 & Q_{xyyz} + \frac{Q_{zzzz}}{2} - \frac{Q_{zzxz}}{2} & 0
\end{bmatrix}$$

$$\begin{aligned}
Q_{xxxx} &= 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} + 2\mathbb{Q}_v^{(1)} + 4\mathbb{Q}_v^{(2)} \\
Q_{xxxz} &= -\mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{yyyy} &= 3\mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} + 2\mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 4\mathbb{Q}_u^{(2)} - 2\mathbb{Q}_v^{(1)} - 4\mathbb{Q}_v^{(2)} \\
Q_{zzzz} &= \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} - 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{zzzz} &= 3\mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} + 2\mathbb{Q}_4^{(1)} + 4\mathbb{Q}_u^{(1)} + 8\mathbb{Q}_u^{(2)} \\
Q_{yyzx} &= \mathbb{Q}_{xz}^{(1)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_y^{\beta(1)} \\
Q_{yyzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} + 3\mathbb{Q}_u^{(3)} - \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)} \\
Q_{zzxx} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(1)} - 3\mathbb{Q}_u^{(3)} + \mathbb{Q}_v^{(1)} + \mathbb{Q}_v^{(3)} \\
Q_{zzxz} &= \mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} + \mathbb{Q}_{xz}^{(3)} + 2\mathbb{Q}_{xz}^{(5)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{xxyy} &= \mathbb{Q}_0^{(1)} + 2\mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_4^{(1)} - 2\mathbb{Q}_u^{(1)} - 2\mathbb{Q}_v^{(3)} \\
Q_{yzzz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} + \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} + 3\mathbb{Q}_u^{(5)} - \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)} \\
Q_{zxxx} &= -\mathbb{Q}_{4y}^{\alpha(1)} + \mathbb{Q}_{xz}^{(1)} + 2\mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(3)} - \mathbb{Q}_y^{\beta(1)} \\
Q_{zzxz} &= \mathbb{Q}_0^{(1)} - \mathbb{Q}_{4u}^{(1)} - \mathbb{Q}_{4v}^{(1)} - \mathbb{Q}_4^{(1)} + \mathbb{Q}_u^{(2)} - 3\mathbb{Q}_u^{(5)} + \mathbb{Q}_v^{(2)} + \mathbb{Q}_v^{(5)} \\
Q_{xyyz} &= \mathbb{Q}_{xz}^{(2)} - \mathbb{Q}_{xz}^{(5)} + 2\mathbb{Q}_y^{\beta(1)}
\end{aligned}$$